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Evaluating the Efficiency of Cytosine and Adenine Base Editors in Correcting Pathogenic *BRCA1* Point Mutations without Double-Stranded DNA Cleavage in Hereditary Breast and Ovarian Cancer (HBOC) Syndrome

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Introduction

The *BRCA1* gene is a tumor suppressor that triggers cell cycle arrest and mediates apoptosis in response to DNA damage such as double-strand breaks (DSBs) [1]. Expression of *BRCA1* is seen in various tissues, particularly playing a crucial role in breast and ovarian tissues [1], [2]. As a result, lifetime susceptibility to breast and/or ovarian cancer is raised, predominantly in women carrying pathogenic *BRCA1* germline mutations [3]. This is known as the Hereditary Breast and Ovarian Cancer (HBOC) syndrome [3]. At present, clinical strategies such as preimplantation genetic testing (PGT-M) or bilateral salpingo-oophorectomy and bilateral mastectomy are utilized to mitigate the risks of breast and ovarian cancer; however, they impose long-term physical and psychosexual side effects with trade-offs in patients' quality of life [4], [5]. In addition, traditional CRISPR/Cas9-mediated genome editing is used as a non-surgical approach relying on the creation of double-stranded breaks (DSBs); consequently, DSB dependence can induce random indels and prevalent embryonic mosaicism, resulting in low efficiency of precise correction [6]. To address these limitations, this study aims to explore the potential of cytosine and adenine base editors, their efficiency, safety and fidelity in the precise correction of *BRCA1* variants through a quantitative meta-analytical approach from existing primary embryonic studies.

Research Questions and Hypotheses

This study intends to explore the potential and current limitations of base editing, specifically using adenine and cytosine base editors, in correcting pathogenic *BRCA1* variants that raise lifetime susceptibility to breast and ovarian cancer. Hence, this study will address the following questions:

1. What is the meta-analytic on-target installation efficiency of cytosine and adenine base editors in pathogenic germline *BRCA1* point mutations?
2. To what extent can cytosine and adenine base editors minimize unwanted byproducts and off-target activity when correcting target *BRCA1* variants?

It is hypothesized that:

- Cytosine and adenine base editors will enable high efficiency in correcting *BRCA1* point mutations allowing successful genome modifications [7]. In comparison to CRISPR/Cas9 -induced double-strand breaks (DSBs), base editor-induced nicks can be corrected more efficiently [8] .
- While base editing generates substantially lower frequency of unwanted double-strand break byproducts such as indels and off-target single-nucleotide variants compared to CRISPR, adenine base editors in particular manifest higher fidelity than cytosine base editors by generating negligible indels and off-target variants consistently [9], [10].

Literature Review

Molecular Pathology of Germline *BRCA1* Point Mutations

The functional RING and BRCT domains are critical in *BRCA1*'s role as a tumor suppressor [11]. To determine the functional consequences of single-nucleotide variants in *BRCA1*, researchers using saturation genome editing (SGE) targeted exons in the N-terminal RING finger domain (encoded by exons 2–7) and the C-terminal BRCT tandem repeats (encoded by exons 16–24). Findings indicate that pathogenic single-nucleotide variants (SNVs) particularly cause missense substitutions and aberrant splicing that disrupts *BRCA1*'s tumor suppressive ability [12]. The RING domains of *BRCA1* act as a molecular scaffold and bind to E2 ubiquitin conjugating enzymes, allowing ubiquitin to be transferred from E2 to substrates [13]. Consequently, when an obligate heterodimer is formed with *BARD1* (*BRCA1*-Associated RING Domain protein 1) through their N-terminal regions, a robust E3 ubiquitin ligase activity is manifested [11]. This activity is vital for double-strand break (DSB) repair through homology-directed repair (HDR) by enhancing *RAD51* recombinase activity and promoting DNA resection [14]. As the RING and BRCT domains have thousands of these pathogenic single-nucleotide variants (SNVs) and variants of uncertain significance (VUS), development of precise genome editing strategies remains critical to correct germline point mutations without dependency on double-strand breaks (DSBs) [12].

Limitations of DSB-Dependent CRISPR/Cas9 Editing

CRISPR/Cas9 editing induces double-strand breaks (DSBs) at specific genomic loci by using a bacterial-derived DNA endonuclease. While Cas9 is an effective tool for DNA repair and target modifications, Cas9 cleavage may lead to chromosomal abnormalities in human embryos such as mosaicism, structural loss of heterozygosity, and chromosomal aneuploidies. In addition, off-target editing can occur in different locations of the genome with similar sequences, generating unwanted indels. Recent findings indicate that double-strand break (DSB) repair in human embryos is an extremely slow process, leaving the breaks unrepaired for hours; consequently, cleavage stage embryos primarily at the G2 stage of the first cell cycle contains high levels of mosaicism at both on-target and off-target locations. Thus, the slow kinetics of Cas9-cleavage and repair prior to the first S-phase limits the potential to reduce embryonic mosaicism and prevent aneuploidies for a precise correction [15]. Moreover, another tumor suppressor gene called *p53* is vital in Cas9 editing; when a double-strand break (DSB) is generated by Cas9, *p53* detects the damage and mediates cell cycle arrest or apoptosis in non-cancerous cells. As a result, this brings profound risks such as lowering editing efficiency in healthy cells and creating selection bias which elevates the risk of the healthy selected cells becoming oncogenic [16].

Base Editing Mechanisms & Fidelity

Base editing represents a major technological advancement over CRISPR/Cas9 enabling direct and precise point mutations without creating DSBs thereby substantially reducing random indels, translocations, and large-scale genomic rearrangements. Instead, DNA base editors consist of a catalytically impaired Cas nuclease, which is a Cas9 nickase (nCas9), fused to a single-stranded DNA (ssDNA) deamination enzyme; for enhanced efficiency in eukaryotic cells, the catalytically impaired nuclease produces a nick in the non-edited DNA strand, promoting cells to fix the non-edited strand by using the edited strand as a template [17]. There are two types of DNA base editors: cytosine base editors (CBEs) and adenine base editors (ABEs) which operate through deamination from the target base. Base transitions occur as a result within around 3 billion base pairs [7]: CBEs which use cytidine deaminases (such as APOBEC or AID) generate C→T transitions by converting cytosine to uracil and ABEs which use engineered transfer RNA adenosine deaminases (TadA) generate A→ G transitions by converting adenine to inosine [18]. These transitions mediate all 4 possible transition point mutations (C→T, T→C, A→G, and G→A) with higher efficiency and less off-target modifications, reducing risks of developing other mutations [19]. Although base editors cannot be utilized to repair frameshift mutations resulting from insertions and deletions, their specificity to repairing point mutations aligns with the criteria to correct pathogenic *BRCA1* single-nucleotide variants (SNVs) without DSB-induced genotoxicity [17].

Meta-Analysis Rationale & Embryonic Studies

As the genome editing technologies advance, a rigorous evaluation of their efficiency, safety, and consistency plays a vital role in transitioning base editing to clinical germline applications to reduce off-target modifications and embryonic mosaicism [20]. The relatively small sample sizes and variable rates of off-target edits reported in primary embryonic studies limit their overall predictive power [9]. Consequently, a meta-analytical approach is required to measure the efficiency of base editing and fidelity across existing studies to establish quantitative safety thresholds, before applying base editing to fix oncogenic *BRCA1* variants [21].

Methodology

This research proposal will use quantitative methods to employ a systematic review and evaluate the efficiency, target fidelity, and safety criteria of base editing technologies. As experimental editing of human germline tissue is restricted by ethical guidelines and sample limitations, secondary quantitative data from published open-access embryonic and germline datasets will be utilized.

Data Collection

Studies evaluating cytosine base editors (CBEs) and adenine base editors (ABEs) in mammalian germline *BRCA1* point mutation will be collected through a directed search across ubMed, NCBI ClinVar, bioRxiv, and medRxiv [22], [23], [24] . Primary quantitative sequencing data on point transition mutations will be included when reviewing studies while data relating double-strand break (DSB) dependent Cas9 tests as well as experiments on frameshift mutations that are not relative to base editor chemistry will be excluded.

Metrics for Meta-Analysis

Analyzed data will be logged in standard data tables (Excel/Google Sheets) targeting three core variables:

1. Target Editing Efficiency (%):

Target Editing Efficiency (%) = (corrected alleles ÷ total alleles) × 100

2. Bystander Editing Rate (%):

Calculate the percentage of unintended C→T or A→G conversions within the editing window (positions 4–8).

3. Off-Target Indel Frequency (%):

Off-Target Indel Frequency (%) = (alleles containing indels ÷ total alleles) × 100

Data Analysis

Pooled averages, ranges, and standard deviations will be calculated using descriptive statistics in Excel/Google Sheets to quantify base editing efficiency and target fidelity [25]. The percentage differences will be used as a comparison between base editing metrics and traditional CRISPR/Cas9 baseline parameters to establish quantitative safety criteria for clinical applications [15].

Bioethical Considerations

This study uses secondary quantitative data extracted from open-access publications from sources such as PubMed, ClinVar, bioRxiv, etc [22], [23], [24]. Direct human subject participation is not involved; thus, the study is exempt from Institutional Review Board (IRB) approval under U.S. Department of Health and Human Services guidelines (45 CFR 46.104) to evaluate base editing in *BRCA1* point mutations [26]. Research conduct will strictly give a

transparent report to avoid publication bias and ensure data integrity [27]. In addition, the quantitative method utilized to evaluate base editing efficiency in this study enables proper safety criteria check and fidelity metrics without experimenting on human embryonic tissues [28]. This allows primary safety concerns that directly trigger the principle of non-maleficence in germline gene modification to be addressed for future gene editing guidelines [21].

Conclusion

This study sets up a meta-analysis framework to evaluate the editing efficiency, target fidelity and clinical safety of using adenine base editors (ABEs) and cytosine base editors (CBEs) in making precise corrections of pathogenic germline *BRCA1* variants causing Hereditary Breast and Ovarian Cancer (HBOC) syndrome. By utilizing secondary quantitative sequencing data from open-access publications, ethical constraints regarding direct human germline experiments are bypassed while addressing the principle of non-maleficence. Through calculations of target editing efficiency, bystander editing rate, and off-target indel frequency, the extracted data for base editing will be benchmarked directly against canonical CRISPR/Cas9 standards. Therefore, necessary empirical safety thresholds for the future bioethical governance can be established through this framework for clinical applications.

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